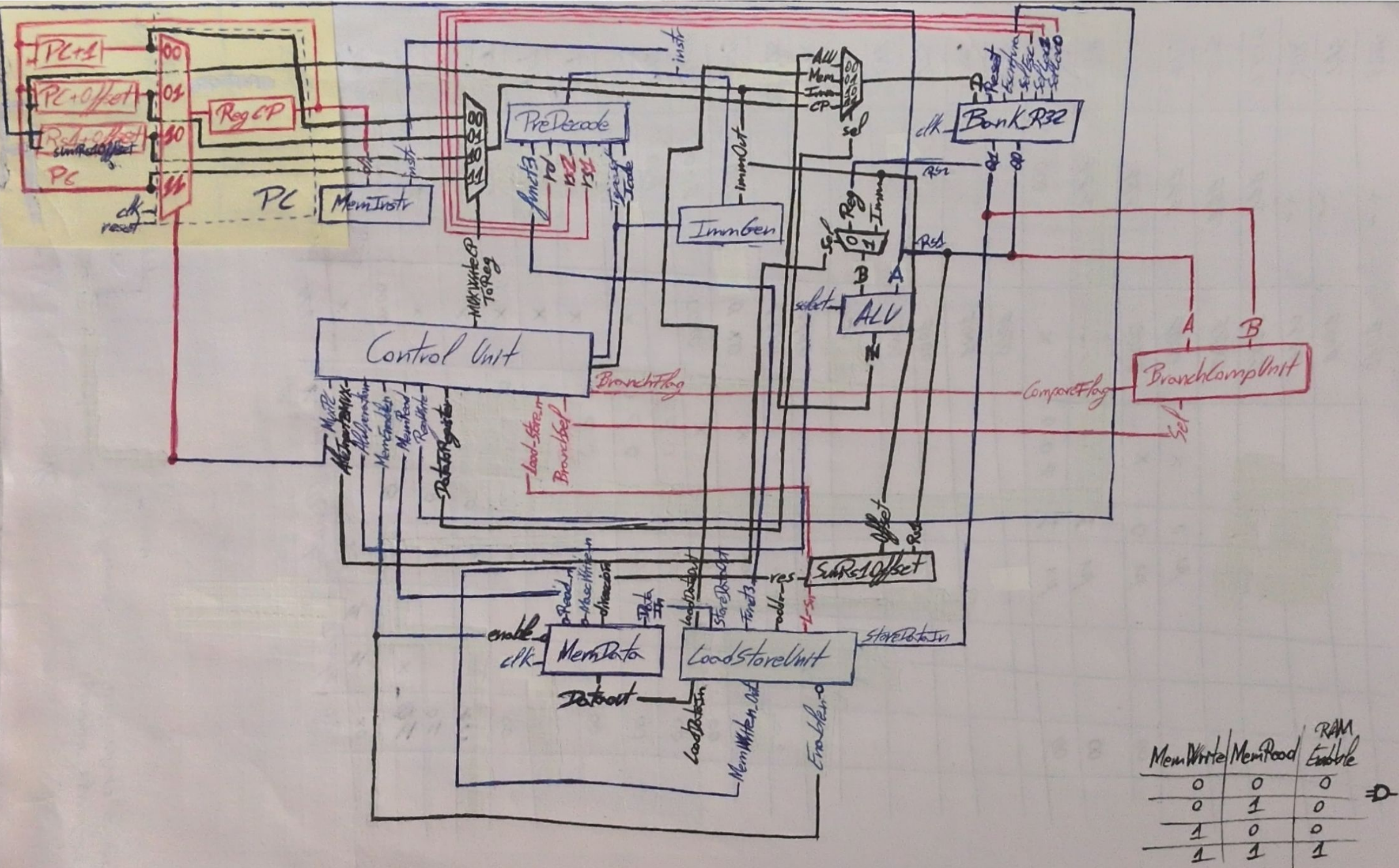
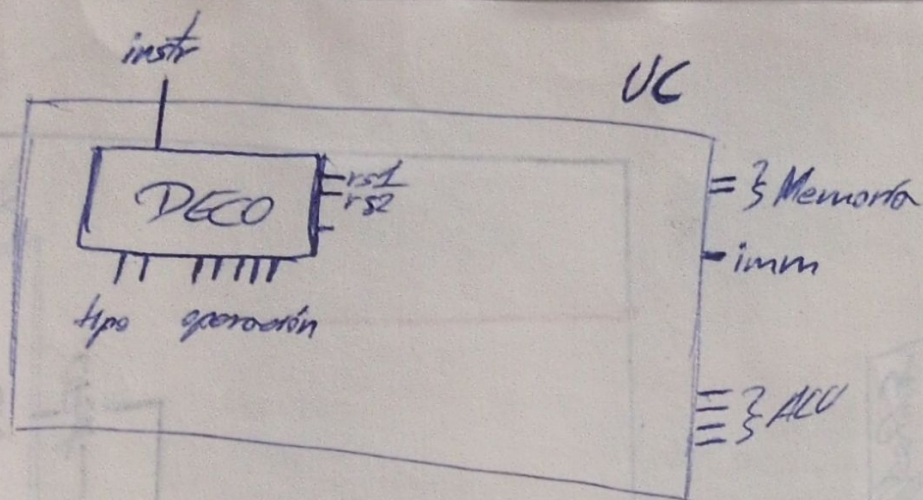




MemWrite	MemRead	RAM Enable
0	0	0
0	1	0
1	0	0
1	1	1



sumador específico
instrucciones de memoria

	Imm Gen	ALU	Memoria			Registros		MUX PC
			E	R	W	Esat	Selecc	
LVI	100	x	x			0	rd	00
ADDI	100	x	x			0	rd	00
JAL	011	x	x			0	rd	01
JALR	001	x	x			0	rd	10
LB	001	x	0	1	0	0	rd	00
LH	001	x	0	1	0	0	rd	00
LW	001	x	0	1	0	0	rd	00
LBU	001	x	0	1	0	0	rd	00
LHU	001	x	0	1	0	0	rd	00
ADDI	001	00x0	x			0	rd	00
SLTI	001	11x0	x			0	rd	00
SLTIU	" "	11x1	" "	" "	" "	" "	" "	" "
XORI	" "	0110	" "	" "	" "	" "	" "	" "
ORI	" "	0101						
ANDI		0100						
SLLI		1000						
SRLI		1001						
SRAI		1010						
SB SH SW	000	x	0	0	1	1	rd	00
ADD	111	00x0	x			0	rd	00
SUB	111	00x1	x			0	rd	00
SLL	111	1000	x			0	rd	
SLT	111	11x0						
SLTU	111	11x1						
XOR	" "	0110						
SRL	" "	1001						
SRA	" "	1010						

Mux With CPT to Reg
Mux CPT to Mux4Reg

00	PC+1
01	PC+Offset
10	Rs1+Offset
11	PC

Mux PC
PC

00	PC+1
01	PC+Offset
10	Rs1+Offset
11	PC

Data to Register
Mux4Reg

00	ALU
01	Mem
10	Imm
11	CP

MemEnable
LoadStoreUnit

1	None
0	Function W/R

MemRead MemEnable
MemData

Enable	Read	
1	X	None
0	0	Lee
0	1	None

BranchSel
BranchCompUnit

00	rs1=rs2 1
01	rs1<rs2 1
10	rs1<=rs2 1
11	'0'

ALU Input Mux
Mux4ALU

0	Reg(rs2)
1	Imm

ALU Operation
ALU32

0000	ADD
0001	SUB
0010	ADD
0011	SUB
0100	AND
0101	OR
0110	XOR
0111	'0'
1000	<<4 SLL
1001	>>4 SRL
1010	>>5 SRA
1011	'0'
1100	SLT
1101	SLTU
1110	SLT
1111	SLTU

RegWControl
RegWrite
RegReadSel
RegReadSel

Bank R32

Enable	Write	Read	Read	
1	X	0-30	31	Lee
0	0-30	0-30	31	Eschibe/Lee

TypeSel	I Code	Inst
S-type	000	0000 SB
		0001 SH
		0010 SW
		1111 Others
I-type	001	0000 ADDI
		0001 ANDI
		0010 ORI
		0011 XORI
		0100 SLLI
		0101 SRLI
		0110 SRAI
		0111 SLTI
		1000 SLTIU
		1001 LB
		1010 LH
		1011 LW
		1100 LBW
		1101 LHW
		1110 JALR
		1111 Others
D-type	010	0000 BEQ
		0001 BNE
		0010 BLT
		0011 BGE
		0100 BLTU
		0101 BGTU
		1111 Others
J-type	011	0000 JAL
U-type	100	0000 LUI
		0001 AUIPC
R-type	101	0000 ADD
		0001 SUB
		0010 AND
		0011 OR
		0100 XOR
		0101 SLL
		0110 SRL
		0111 SRA
		1000 SLT
		1001 SLTU
		1111 Others
System	111	0000 WFI
		1111 Others

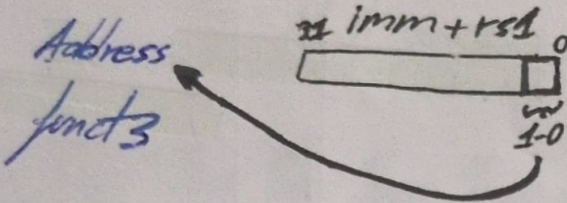
LSU

Enable-n
Write-n
Read-n

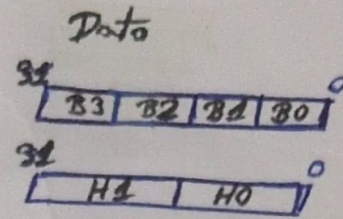
StoreDataIn ← se trunca el dato
StoreDataOut

LoadDataIn

LoadDataOut ← se expande

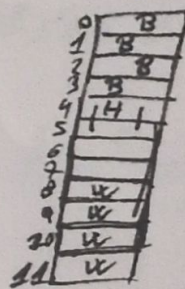
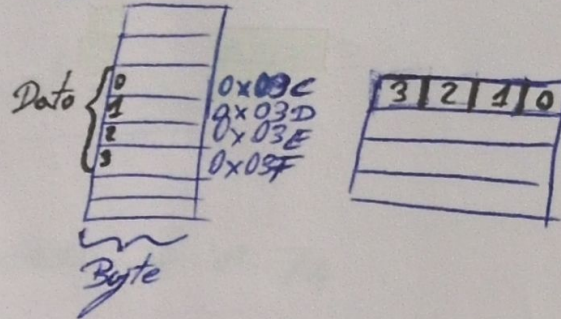


func3
000 B
001 H
010 W
100 BU
101 HV



B/H	Addr	
0	00 01 10 11	B0 B1 B2 B3
1	00 01 10 11	H0 H1

Byte-Addressed

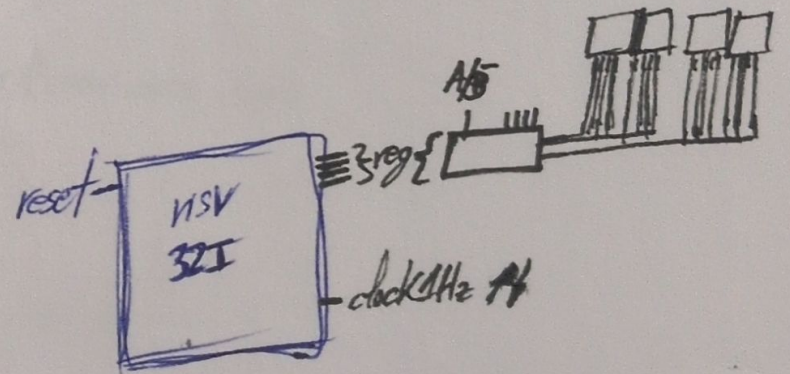
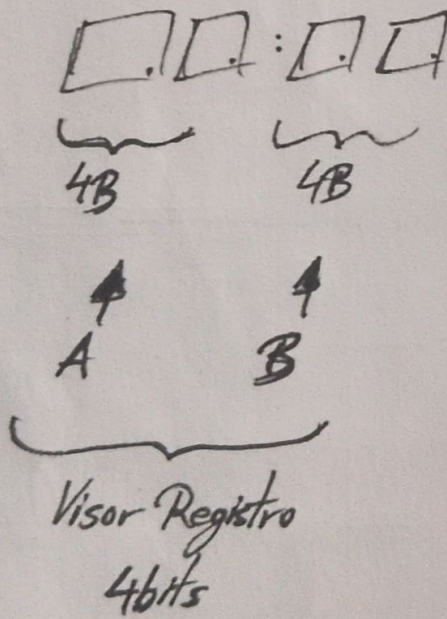


Barra
H múltiplos 2
W múltiplos 4

Almacén

Se asume alineamiento para no complicar la lógica, además de no implementar lanzamiento de excepciones.

Display



³¹ 28 26 25 24 20 19 15 14 12 11 7 6 0
0000 0000 0011 1010 0000 0010 0001 0010 0011 5-type
 11-5 3 0 2 14-0 2 3
 0 0 offset 0000 0000 0010 (2) ~~RRR RRR RRR RRR~~ (2)

³¹ 28 27 23 19 15 11 7 6 4 3 0
0000 0000 1010 0000 0000 0000 1001 0011 I-type
 11-0
 offset 0000 0000 1010 (a)

³¹ 28 24 23 19 15 11 7 4 3 0
¹² 1111 1110 0000 0000 0000 1110 1110 0011 B-type
 10-5 4:1 11
 offset 11 111111 1110 & '0' /4

11 111111 111 (~~FFF~~) X ~~FFFF FFFF~~

³¹ 28 23 19 15 11 7 3 0
²⁰ 0000 0000 1000 0000 0000 0000 0110 1111 J-type
 10-1 11 19-12

offset 0 0000 0000 0 000 0000 100 & '0' /4
 0... 010 (2)

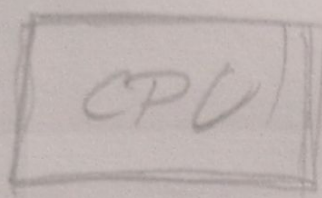
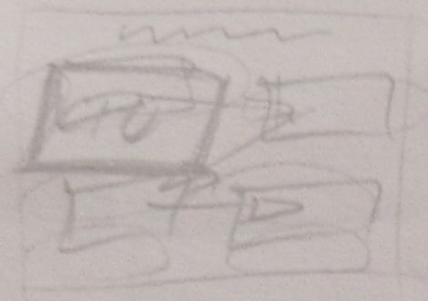
³¹ 28 23 19 15 11 7 3 0
0001 0010 0011 0100 0101 0000 0011 0111 V-type
 31-12

0001 0010 0011 0100 0101 & 0000 0000 0000
 (12345000)

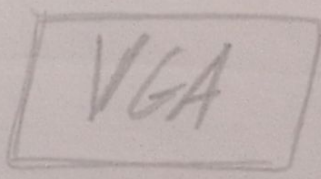
ALU32

Cod.		
0000	Arith	ADD
0001		SVB
0010		ADD
0011		SVB
0100		AND
0101	Logic	OR
0110		XOR
0111		'0'
1000		'0'
1001	Shift	<< assigned SLL
1010		>> M SRL
1011		>> S SRA
1100		'0'
1101	Comp	SLT
1110		SLTU
1111		SLT
1111		SLTU

A < B



- CP
- MemInst
- ProtCode
- ImmGen
- UC
- ALU
- Branch Reg
- MemData
- LoadStoreUnit



- AuxRAM
- AuxReg
- BinToDec
- DecToDigit
- TextRAM
- Vga-control
- ROM-letter

